

FINAL PROJECT

FDA Product Recall Analysis Using Normalized SQL Tables from XML and CSV Data Sources

Abstract

This document has details about my last-minute change and the updates highlighted by Professor Polnar. You can find the updated Entity Relationship Diagram, the changed normalized tables, and the reason behind the different steps and tables.

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FDA RECALL

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1. Introduction

This project focuses on building a high-performance, normalized SQL database to analyze FDA product recall data spanning multiple years. The dataset originates from official FDA sources in **XML** and **CSV** formats, covering recalls from **2018 to 2023**. The objective is to extract, clean, normalize, and insert this complex real-world data into structured relational tables that support advanced queries, historical analysis, and downstream data warehousing operations such as Slowly Changing Dimensions (SCDs).

The system leverages **Python** for ETL (Extract, Transform, Load) processes and **MySQL** for structured storage, allowing efficient querying and scalability for future integration with tools like **Power BI** or **Tableau** for visualization.

2. Objectives

- To ingest multi-format FDA recall data (XML and CSV) into a structured SQL database.
- To design and normalize relational schemas for product, company, recall, and metadata.
- To implement efficient ETL scripts in Python for fast and accurate data insertion.
- To ensure data consistency through foreign key constraints and de-duplication.
- To support **Slowly Changing Dimension (SCD Type 2)** tracking of company name changes.
- To prepare the system for future analytical and visualization layers.

3. Tools and Technologies

Component	Tool / Language
Database	MySQL
ETL Scripting	Python 3.13
Parsing XML	ElementTree
Parsing CSV	Pandas
Date Handling	dateutil.parser, datetime
Data Modeling	SQL Normalization (3NF), BCNF
Deployment	Windows Environment

4. Data Sources

- **XML Files:** 6 annual recall datasets (`recalls2018.xml` to `recalls2023.xml`) containing structured recall notices.
- **CSV File:** `updated_fda_product_recalls.csv` containing product, company, classification, and distribution details.

5. Schema Design Overview

The project separates XML and CSV records into two parallel normalized schemas:

- **XML Schema Tables:** XML_PRODUCT, XML_COMPANY, XML_REASON, XML_SOURCE_URL, XML_RECALL_BRIDGE, etc.
- **CSV Schema Tables:** CSV_PRODUCT, CSV_COMPANY, CSV_REASON_RECALL_DETAILS, CSV_RECALL_BRIDGE, etc.
- A **merged schema** (FDA_* tables) is prepared for analytical use and SCD handling.

All relationships are enforced using **foreign key constraints**, and bridge tables like XML_RECALL_BRIDGE and CSV_RECALL_BRIDGE serve as fact tables that connect dimensional attributes.

6. ETL Strategy

Two Python scripts were developed:

- **XML Parser & Inserter:** Parses XML using ElementTree, extracts key fields, checks for duplicates, inserts new records, and populates bridge tables.
- **CSV Loader:** Reads and cleans CSV using pandas, auto-parses unformatted dates, handles missing values, maps dimensions to IDs, and inserts into normalized tables.

Optimizations:

- Use of INSERT IGNORE and SELECT to avoid duplicate entries.
- Caching and foreign key indexing for fast lookups.
- Batch processing (future scope).

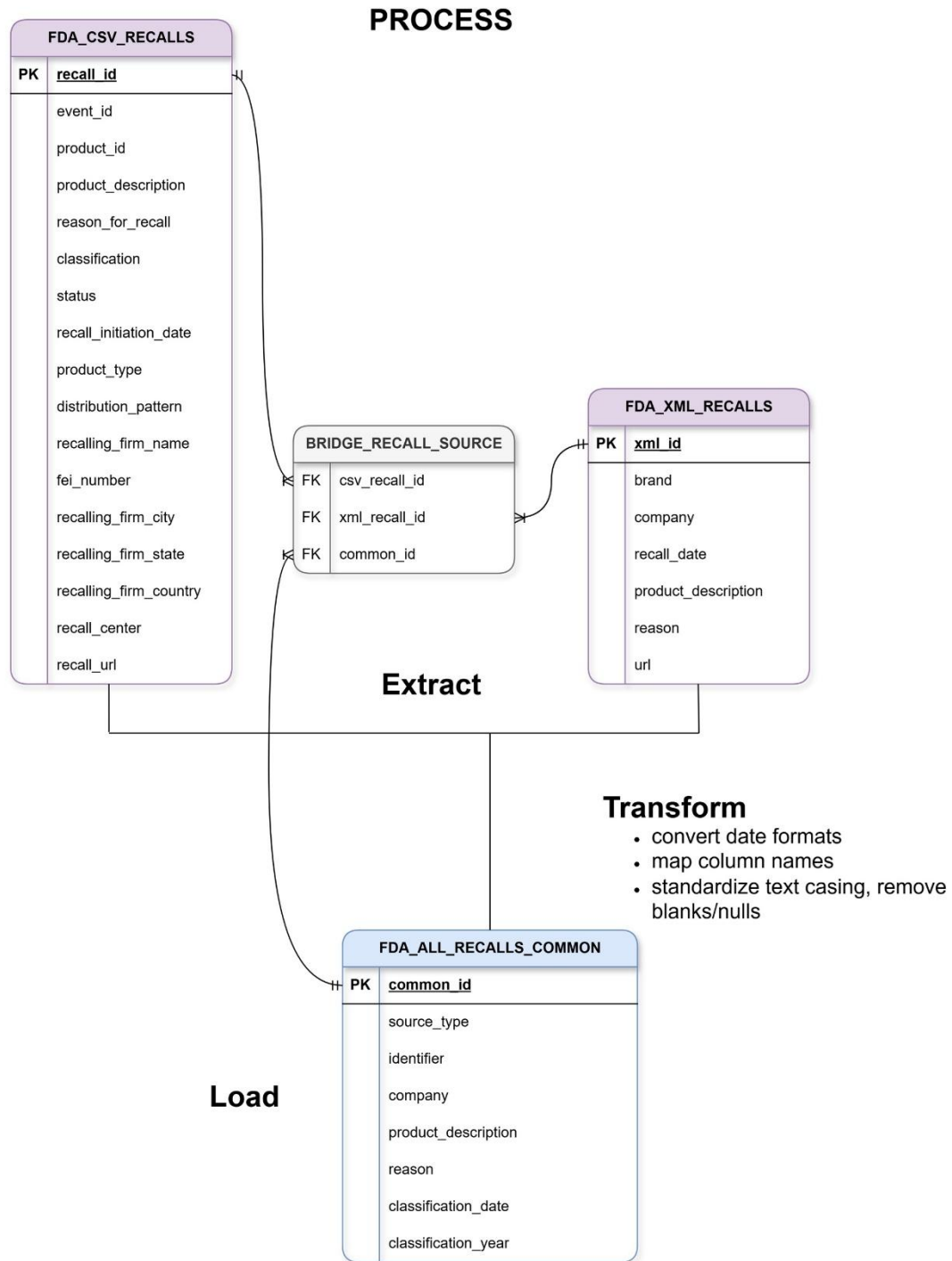


Fig: ETL Process

7. Entity Relationship Diagram

1) Initial Design → Redone Design

I have changed the initial design into this updated diagram. Since my product table was over-normalized, I combined product_class and product_status in the csv_product table. But, product_type stays separate from the product table to avoid anomalies.

Similarly, I also combined the REASON and RECALL_DETAILS tables, since creating a new reason for the same recall is quite unlikely.

8. What happens if the company name is not changed?

1) When nothing is changed, which part of the code handles it?

```
INSERT INTO DIM_COMPANY (...)  
SELECT ...  
FROM FDA_COMPANY f  
LEFT JOIN DIM_COMPANY d  
  ON f.COMPANY_ID = d.COMPANY_ID  
  AND d.IS_CURRENT = 1  
WHERE d.COMPANY_ID IS NULL  
  OR d.COMPANY_NAME <> f.COMPANY_NAME;
```

Step 1: No rows updated

Step 2: No rows inserted

1. Existing DIM_COMPANY record remains with IS_CURRENT = 1 and VALID_TO = '9999-12-31'
2. That record continues to represent the current company version

9. STAR Schema

The FDA Recall Snapshot Star Schema is a dimensional data model designed to support efficient analytical querying of product recall data. At its core is the FACT_RECALL_SNAPSHOT table, which captures key recall metrics such as recall_count, classification date, and snapshot date, and links them to various descriptive dimensions. Each foreign key in the fact table points to a corresponding dimension table—DIM_COMPANY, DIM_PRODUCT, DIM_PRODUCT_TYPE, DIM_REASON, DIM_RECALL_SOURCE, DIM_IDENTIFICATION, and DIM_DATE—providing rich context for slicing and dicing recall data.

This schema incorporates Slowly Changing Dimensions (SCDs) to support historical analysis. SCD Type 2 is applied to DIM_COMPANY and DIM_IDENTIFICATION to track changes over time (e.g., name changes or metadata updates), while other dimensions use SCD Type 1 or 0 for simpler overwrites or static data. The model ensures a clean separation of facts and dimensions, making it highly suitable for BI tools and regulatory reporting, enabling users to answer complex questions like trends over time, recall reasons by product type, or company recall histories.

10. Text Vectorization for Natural Language Queries

To enhance the analytical capabilities of the FDA recall system, a natural language search feature was implemented using **cosine similarity** with **TF-IDF (Term Frequency–Inverse Document Frequency)** vectorization. This allows users to input free-text queries and retrieve the most relevant recall records based on textual similarity. Key attributes from the normalized SQL tables—including company name, product description, recall reason, product type, and source—were combined into a unified combined_text field for each record. This text was transformed into numerical vectors using TfidfVectorizer, which weighs terms by their frequency in a document relative to their occurrence across the entire dataset.

When a user enters a search prompt, it is similarly vectorized and compared against all existing recall records using **cosine similarity**, which measures the angle between the user vector and each record vector. The closer the vectors (angle near zero), the higher the similarity score. The top-N most similar records are returned in descending order of relevance, allowing intuitive semantic search without requiring exact keyword matches. This approach significantly improves data accessibility, especially for non-technical users or exploratory analysis, and lays the foundation for integrating more advanced NLP techniques in the future.

11. Conclusion

This project provides a comprehensive, scalable backend foundation for FDA recall analysis. By combining strong database design, robust Python scripting, and regulatory-quality data handling, the system is capable of supporting compliance reporting, root cause investigation, and visualization dashboards for health data analysts, manufacturers, and public safety researchers.

Appendices

STAR SCHEMA

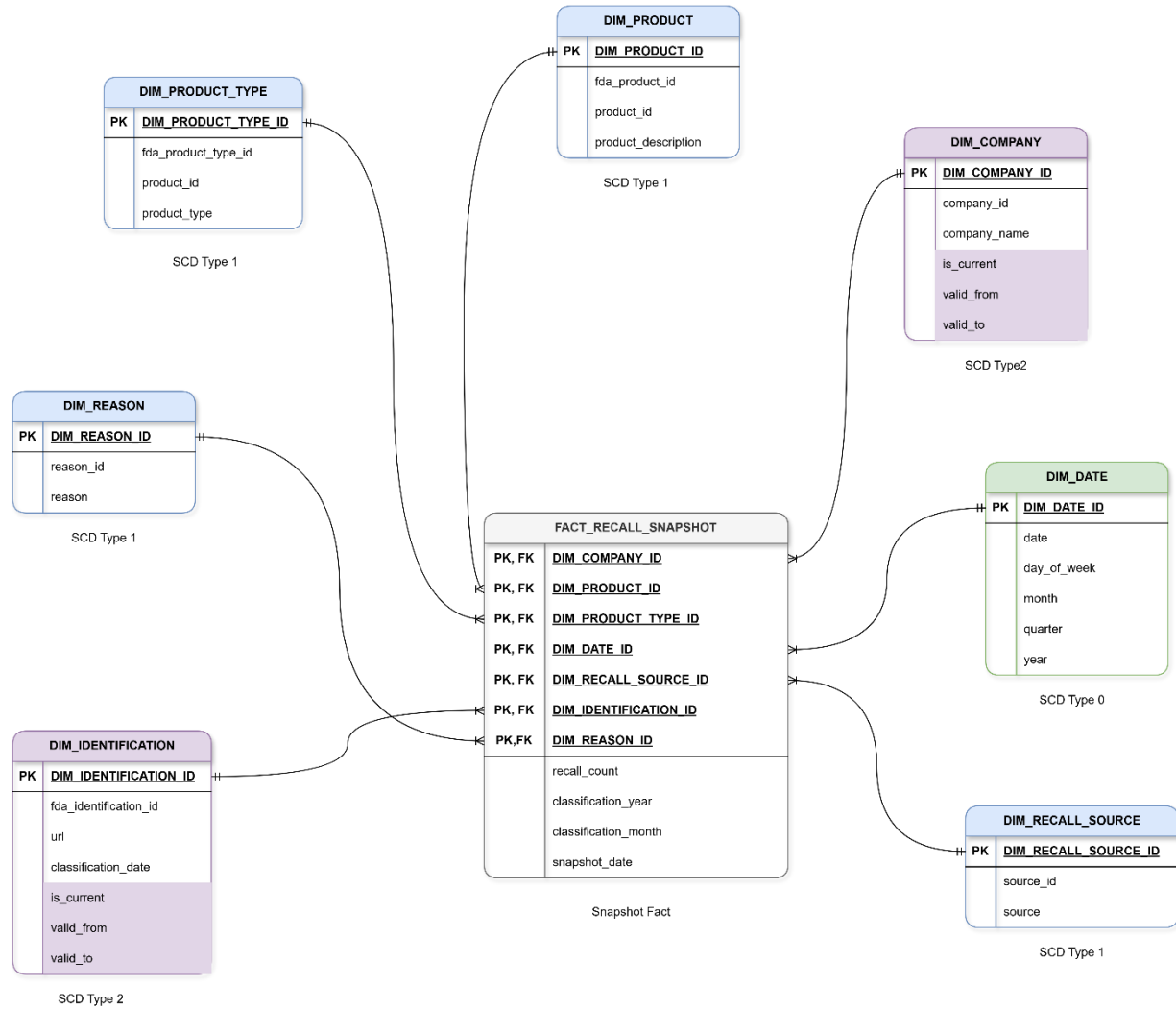


Fig.1 STAR Schema

Entity Relationship Diagram

legend

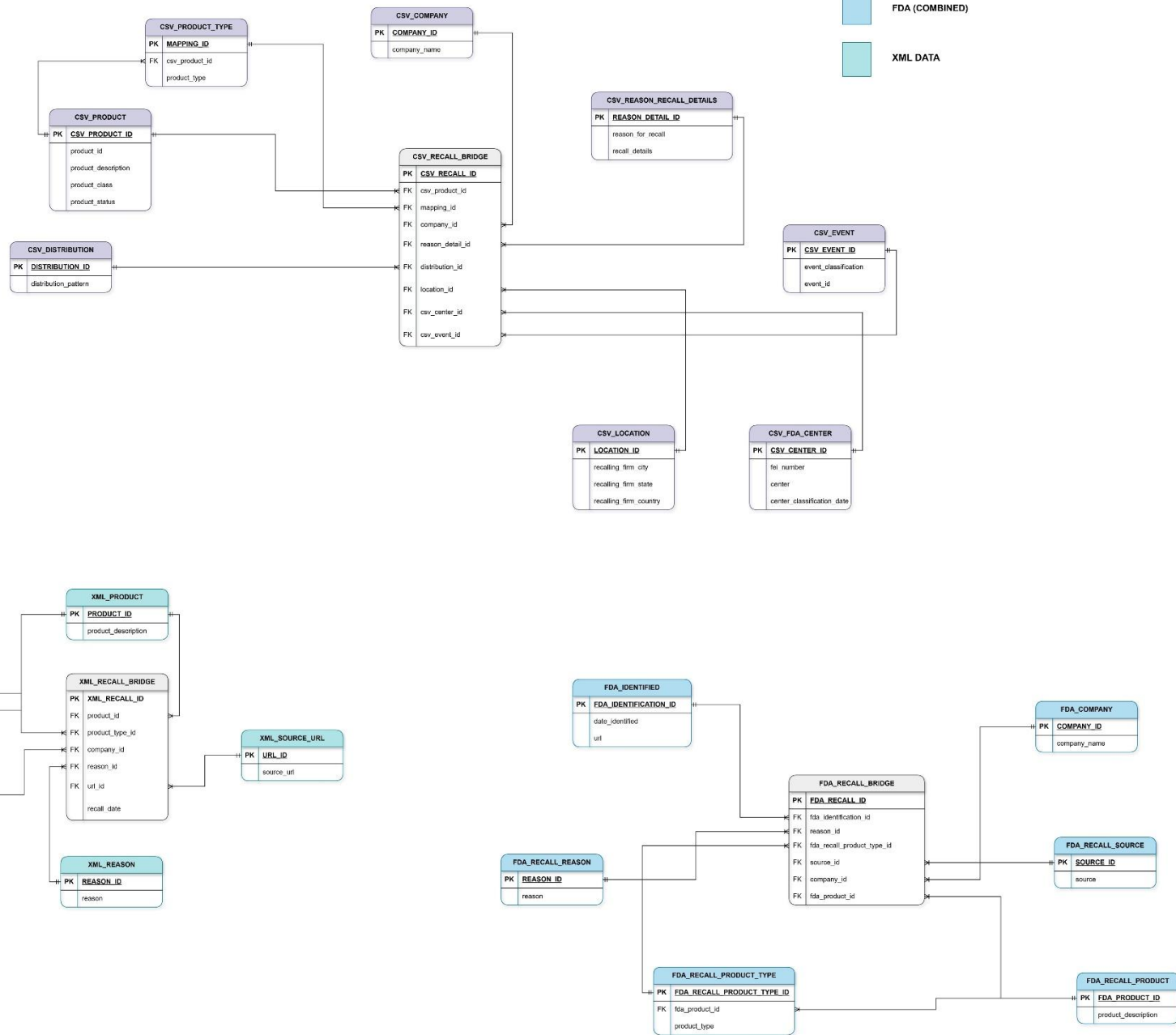
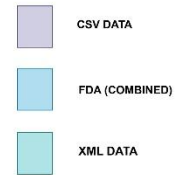


Fig.2 Entity Relationship Diagram (Redone)

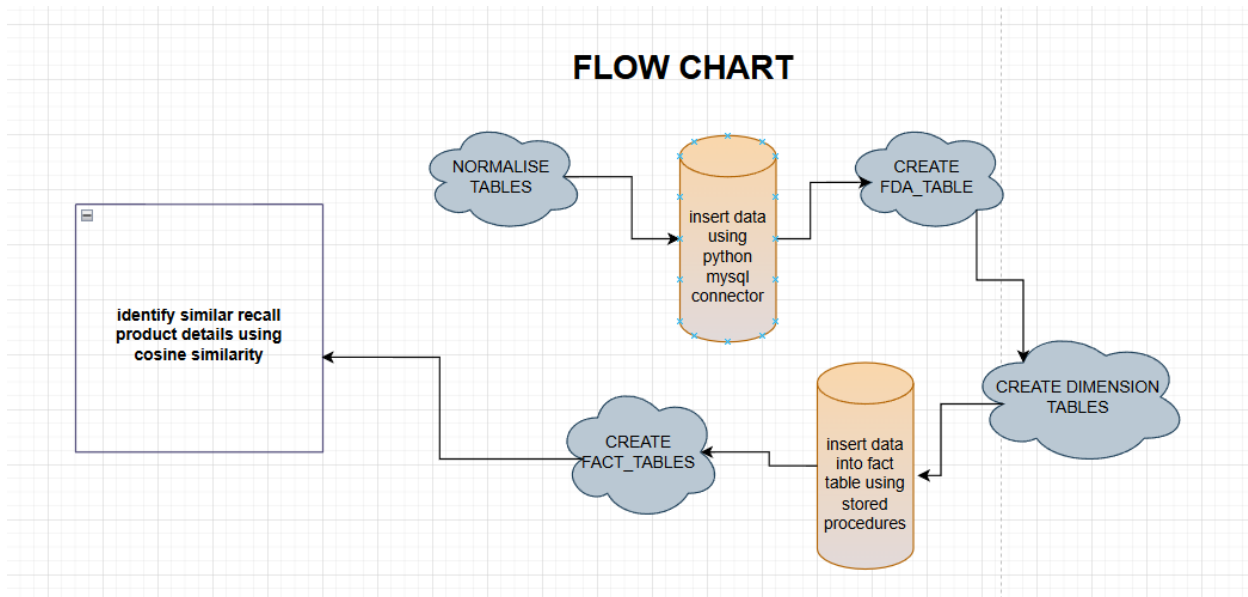


Fig.3 Flow Chart

```

PS C:\Users\Shwetha\Documents\ADB Project> & C:/Python313/python.exe "c:/Users/Shwetha/Documents/ADB Project/python/just_to_find_unique_names.py"
['Biologics', 'Devices', 'Drugs', 'Food/Cosmetics', 'Tobacco', 'Veterinary']
PS C:\Users\Shwetha\Documents\ADB Project>

```

Fig.4 Output showing Product types

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\Shwetha\Documents\ADB Project> & C:/Python313/python.exe "c:/Users/Shwetha/Documents/ADB Project/python/user_prompt_similarity.py"
Connected to MySQL database!
Enter your search query: nut allergy

Top Matches:

```

	COMPANY_NAME	REASON	SOURCE	similarity_score
14558	Family Dollar Stores, LLC.	CGMP Deviations: product held outside appropri...	CSV	0.355623
905	Midwest Northern Nut	Undeclared peanuts and various tree nuts	XML	0.338722
9911	Texas Star Nut and Food Co., Inc.	One case (out of 4332 cases) of pine nuts was ...	CSV	0.281338
640	Oskri	Listeria monocytogenes	XML	0.259480
10862	Dr. Reddy's Laboratories, Inc.	Failed dissolution specifications	CSV	0.245584
7688	King Bio Inc.	Microbial Contamination	CSV	0.234315
649	Thrive Market	Listeria	XML	0.230530
14951	Lemonland Food Corp	Seasoned Nut was recalled due to undeclared Pe...	CSV	0.227901
277	Healthy Nut Factory	May contain Salmonella	XML	0.217709
589	Wholesome Pantry	Potential to be contaminated with Listeria mon...	XML	0.216743
934	Healthy Nut Factory	potential to be contaminated with Salmonella	XML	0.215393
1676	Allergy Bee Gone for Kids	Product contains elevated levels of Bacillus c...	XML	0.212338
620	Joy	Allergy Alert on Undeclared Sulfites	XML	0.210661
43	No brand name; "PACKED BY FIRST SOURCE" printe...	Undeclared pecan and cashew	XML	0.207436
246	Eataly	Undeclared tree nut (walnut)	XML	0.203334

```

[15 rows x 6 columns]
PS C:\Users\Shwetha\Documents\ADB Project>

```

Fig.5 Output showing similar recall details